

What actually moves boundary error in echocardiographic segmentation?

A reproducibility study of real-time instance segmentation on CAMUS

[TODO: Author names]^a

^a[TODO: Affiliation]

Abstract

Deep segmentation networks report mean Dice above 0.90 on the CAMUS echocardiography benchmark, at or beyond the level at which two cardiologists agree with one another. We ask what, if anything, still improves boundary accuracy in that regime, using real-time instance segmentation as the test bed.

Applying a detector to CAMUS first requires resolving a representational obstacle: the myocardium is an annulus, and an instance mask is supervised as a single closed polygon, which has no hole. We encode the epicardium as a filled contour and recover the myocardium by subtraction, and quantify what any such encoding can express as an explicit *ceiling* — a perfect model under it attains 0.997 Dice at 0.308 mm HD95, exactly one pixel.

We then run thirteen factors over roughly fifty training runs on a single consumer GPU: encoding, vertex budget, model scale and generation, input resolution, mask-prototype resolution, augmentation, pretraining, training-set size, a boundary-weighted loss, the loss gains, and the optimiser. Critically, we first measure run-to-run variance. Three seeds of one configuration span 0.134 mm HD95, and a paired Wilcoxon test over the 200-frame test set calls that difference significant ($p=0.035$). Read against that floor, *no factor improves on a stock configuration*. What does register is training-side: removing augmentation costs 0.318 mm, and COCO initialisation is worth 0.183 mm.

Averaged over three seeds the resulting model reaches 3.809 mm HD95 at 2.7 M parameters — the best in the benchmark, ahead of the strongest baseline by 0.271 mm, twice the measured seed floor — with mean Dice 0.9113, which is statistically indistinguishable from the four leading baselines. It is near-invariant to input resolution where encoder-decoders scale 4–8× with

pixel count.

Three findings generalise beyond this architecture. Overlap has saturated on CAMUS, so boundary distance in millimetres is what discriminates methods. Single-seed ablations on this benchmark produce false positives. And the reference implementation for CAMUS ejection fraction selects a contour by position rather than area, which is harmless on ground truth but drives EF correlation from $r=0.911$ to $r=0.030$ on model predictions. All code, the machine-readable experiment plan and every run artefact are released.

Keywords: Echocardiography, cardiac segmentation, reproducibility, benchmarking, instance segmentation, ejection fraction

1. Introduction

Two-dimensional echocardiography is the first-line modality for assessing cardiac function. Delineation of the left-ventricular (LV) endocardium, myocardium and left atrium underpins every downstream measurement, ejection fraction (EF) chief among them, and automating it has been an active target since the release of the CAMUS benchmark [1]. Progress has been rapid: encoder–decoder networks [2, 3] and transformer hybrids [4, 5] now routinely report mean Dice above 0.90.

That number deserves scrutiny. The authors of CAMUS report an average inter-observer Dice of 0.899 on this data [1]: two cardiologists tracing the same frames agree with one another to about that level. A model scoring 0.92 is therefore not 0.02 away from being correct; it is already past the point where the reference itself is stable, and further gains in overlap increasingly measure conformity to one annotator’s habits rather than anatomical accuracy.

This paper asks what still moves accuracy in that regime, and answers it mostly in the negative. We use real-time instance segmentation as the test bed, for two reasons. First, it is a genuinely different architectural family from the encoder–decoders that occupy the benchmark, so if the ceiling is a property of the task rather than of one architecture, a detector should hit it too. Second, it is the deployment-relevant one: handheld probes push echocardiography to the bedside, where inference must run on an embedded accelerator alongside a live image stream.

Applying a detector to CAMUS is not, however, a matter of changing the dataloader. An instance mask is supervised as a *single closed polygon*, and one of the three CAMUS structures — the myocardium — is an annulus. A polygon has no hole. It is natural to assume that how one

resolves this matters, and we show that it does not, along with almost everything else one might tune.

1.1. Contributions

An encoding for nested cardiac structures, and its ceiling. We define the second class as the *filled* epicardial disc and recover the myocardium by subtraction, keeping the mask head on simply-connected shapes (Sec. 3.2). We quantify what such an encoding can express by converting ground truth to polygons, rasterising it back and scoring it against the untouched reference. At 96 vertices this reaches 0.997 Dice and 0.308 mm HD95 — exactly one pixel at the CAMUS spacing. The polygon representation is therefore not what limits a detector on this task.

A measured noise floor, and ablations read against it. Before interpreting any comparison we train one configuration under three seeds. The spread is 0.134 mm HD95, and a paired Wilcoxon test over the 200 test frames calls two such runs significantly different ($p=0.035$). We therefore treat 0.134 mm as the smallest credible effect. This is not a formality: several of the differences we would otherwise have reported as significant, and several reported in the literature, are smaller than this.

Thirteen factors, no improvements. We sweep encoding, vertex budget, model scale and generation, input resolution, mask-prototype resolution, augmentation recipe, pretraining, training-set size, a boundary-weighted loss, the three loss gains and the optimiser — roughly fifty training runs on one laptop GPU (Sec. 4). None improves on a stock configuration beyond the noise floor. Higher-resolution mask prototypes are significantly *worse*. What does register is training-side: augmentation and initialisation.

A like-for-like comparison, including latency. We evaluate on the official CAMUS split, verified byte-identical to the released subgroup lists, against nine semantic-segmentation baselines under one protocol, with HD95 at native resolution in millimetres. Because published latencies are measured on heterogeneous hardware, we re-time every model ourselves under a single protocol, at two input resolutions, which shows the detector to be near-invariant to resolution where encoder-decoders scale with pixel count.

A fault in the reference EF implementation. The official CAMUS ejection-fraction code selects a contour by position rather than by area. This is invisible on ground truth, which is single-component, but model predictions carry occasional spurious blobs: on one baseline, 6%

of frames, which was sufficient to drive EF correlation from $r=0.825$ to $r=0.080$ (Sec. 4.7). Any EF computed from predictions with the reference code at native resolution is unreliable without a connected-component filter.

Our aim is not a leaderboard position. It is to establish, with the variance measured rather than assumed, which of the choices practitioners routinely tune actually matter on a benchmark that has reached annotator agreement — and to document the measurement pitfalls we encountered, several of which affect published numbers.

2. Related work

2.1. Segmentation on CAMUS

CAMUS [1] provides apical two- and four-chamber views for 500 patients with expert delineations of the LV endocardium, LV myocardium and left atrium at end-diastole (ED) and end-systole (ES), together with an official patient-level split. The original study established that a plain U-Net trained on a sufficiently large annotated set outperforms classical methods and lands, on average, below inter-observer variability though above intra-observer variability.

Subsequent work has largely followed the encoder–decoder line: nnU-Net’s self-configuring pipeline [3], transformer hybrids such as TransUNet [4] and Swin-UNet [5], and more recently state-space and Kolmogorov–Arnold variants. Reported mean Dice on this benchmark now clusters between 0.90 and 0.95. Two observations motivate the present study. First, that band straddles the 0.899 inter-observer figure reported by the dataset’s own authors, so differences within it are difficult to interpret as anatomical accuracy. Second, comparisons across papers are frequently not like-for-like: protocols differ in the split used, in whether boundary metrics are computed on a resampled grid or at native resolution, and in the hardware on which latency is measured.

2.2. Anatomically constrained and boundary-aware segmentation

Because overlap saturates, several lines of work target the boundary or the shape prior directly. Post-hoc anatomical correction enforces topological validity on network outputs without retraining. Contour-based and curve-deforming formulations regress vertex positions rather than dense masks. Boundary-weighted and distance-transform losses reweight the objective toward the contour. We test the last of these directly (Sec. 4) and find it does not help on this benchmark, which we attribute to the contour being precisely where the reference labels are least reliable.

2.3. Real-time instance segmentation

Detector-based instance segmentation descends from Mask R-CNN [6] and, for the real-time branch, from YOLACT’s prototype-mask formulation [7]: the network predicts a small basis of mask prototypes plus per-instance coefficients and composes them at a fraction of the input resolution. The YOLO family [8] carries this head in its modern versions and is designed for the latency regime that bedside echocardiography requires.

The formulation carries an assumption that does not hold for cardiac anatomy: an instance is a single closed polygon. Nested and annular structures — of which the myocardium is the canonical medical example — have no such representation. To our knowledge this obstacle has not been examined on CAMUS, and detector-based segmentation has not been placed on the same measurement footing as the semantic-segmentation literature.

2.4. Reproducibility and evaluation practice

A recurring finding in empirical machine learning is that reported improvements frequently fall within the variance induced by random seeds, and that comparisons are sensitive to evaluation details that papers do not always specify. Medical imaging inherits this problem with an additional complication: test sets are small — 50 patients here — so paired significance tests over a few hundred frames are powerful enough to certify differences that a repeat run would erase. Our measurement of the seed floor (Sec. 4.1) quantifies this for CAMUS specifically, and our audit of the reference EF implementation (Sec. 4.7) shows that an evaluation fault can survive in widely used released code precisely because it is invisible on ground truth.

3. Materials and methods

3.1. Dataset and splits

CAMUS [1] contains apical two-chamber (2CH) and four-chamber (4CH) views for 500 patients, annotated at ED and ES for three structures: LV endocardium ($\ell=1$), LV myocardium ($\ell=2$) and left atrium ($\ell=3$). We use the official 400/50/50 patient-level split. We verified it byte-identical to the `subgroup_training`, `subgroup_validation` and `subgroup_testing` lists released with the dataset, so results here are directly comparable with published work that uses the official protocol. This yields $400 \times 2 \times 2 = 1600$ training and 200 test frames.

Each patient additionally carries a fully annotated half cardiac cycle. Exported at every annotated frame this gives 17,006 training images, used in Sec. 4 as a data-scaling factor. We confirmed that no test or validation patient appears in that expansion.

Images are exported at native resolution with per-image min–max normalisation. No resampling to a common grid occurs at any stage of dataset construction, so the pixel spacing recorded in the NIFTI header — typically 0.308 mm — remains valid for boundary metrics.

3.2. The annulus problem

The myocardium is the ring between the endocardial and epicardial contours: an annulus, and therefore not representable as the single closed polygon that an instance-segmentation label must be. Two encodings exist.

Filled-contour decomposition. We define the second trained class as the *epicardial disc*, $\ell \in \{1, 2\}$, which is simply connected. All three trained classes are then blob-shaped and the mask head never has to represent a hole. At inference the CAMUS label map is recomposed by painting the predicted disc as myocardium and overwriting its interior with the predicted endocardium,

$$\hat{M}_{\text{myo}} = \hat{M}_{\text{epi}} \setminus \hat{M}_{\text{endo}}, \quad (1)$$

so the hole emerges from the interaction of two predictions rather than from a single mask.

Seam-cut annulus. Alternatively the ring is written as one polygon that traverses the outer contour, crosses to the inner contour along a zero-width seam, traverses it in the opposite winding, and returns. Scan-line filling honours this and yields a genuine annulus; we verified that the rasterisation routine used to build training targets reproduces the ring at 0.994 Dice, and not the filled disc (0.49–0.79).

We note that a third apparent option — taking only the *outer* contour of the ring — is not a distinct encoding. The outer boundary of the myocardial ring *is* the epicardial contour, making it identical to the filled decomposition.

3.3. Encoding ceiling

Both encodings are lossy at a finite vertex budget, so we measure their ceiling directly rather than assume it is negligible. Ground-truth label maps are converted to polygons exactly as the training targets are, rasterised back, and scored against the untouched native ground truth. This

bounds any model trained on that representation. We report two quantities: the Dice of the recomposed four-class map, which is what evaluation ultimately sees, and the Dice of the raw second-class target, which is what the mask head is asked to reproduce. The second is the discriminating one, because Eq. 1 re-carves the hole under either encoding and therefore conceals the difference.

3.4. Evaluation protocol

To keep our numbers comparable with the baselines evaluated on this split we reproduce their protocol rather than improve on it:

- Overlap metrics (Dice, IoU) are computed per frame and per class on a 256×256 grid with a smoothing constant of 10^{-6} .
- Boundary metrics (HD95, ASSD) are computed at *native* resolution using each patient’s true pixel spacing, and are reported in millimetres. Computing them on a resampled grid while retaining the native spacing under-reports distances by the resampling factor.
- Predicted instance masks are resampled to native resolution by nearest neighbour before scoring; for each class the single highest-confidence instance is kept.

We verified this reimplementaion against the reference metric code on perturbed ground truth: HD95 agrees to 2×10^{-12} mm and ASSD exactly. Our implementation streams the pairwise boundary-distance computation in chunks rather than materialising the full distance matrix, which keeps native-resolution evaluation within the memory budget of the same laptop.

3.5. Latency protocol

Published latencies for the baselines were measured on other hardware and cannot be mixed with ours. We therefore re-time every model under a single protocol: batch size 1, CUDA-synchronised around each iteration, ten warm-up iterations discarded, median of fifty timed iterations reported (the median rather than the mean, because scheduler noise on a laptop skews the latter).

Three quantities are separated, because input resolution, architecture and framework overhead are otherwise confounded. *Raw forward* is the bare module forward pass with no pre- or post-processing, measured for every model at both 256 and 640 px so that resolution is isolated

from architecture; this requires no retraining and no accuracy is claimed at the off-design size. *End-to-end* is the full inference call including letterboxing and native-resolution mask assembly. *ONNX* is the exported graph under ONNX Runtime’s CUDA provider.

3.6. Ejection fraction

EF is defined across the pair of cardiac phases, $EF = (EDV - ESV)/EDV$, and therefore requires pairing each patient’s ED and ES predictions in both views. Volumes are computed with Simpson’s biplane method of disks using the official CAMUS implementation, at native resolution with each view’s true spacing. We verified our copy of that code bit-identical to the reference notebook published with the dataset: across all 50 test patients the maximum absolute difference in both EDV and EF is 0.0000.

We additionally compute an *oracle*: EF derived from the ground-truth masks through the identical pipeline. This separates the error contributed by segmentation from the irreducible gap between mask-derived EF and the clinically recorded value, and it is the reference point against which model EF should be read.

3.7. Training configuration

Unless stated otherwise we train YOLO26n-seg initialised from COCO weights at 640 px, batch 8, for 100 epochs with early stopping (patience 30), on a single NVIDIA RTX 4060 Laptop GPU (8 GB). YOLO26 is the newest generation with published segmentation weights and is end-to-end: it predicts a fixed instance set with no non-maximum suppression, which removes a data-dependent stage from the latency budget.

We use the framework’s automatic optimiser selection, which for this workload chooses MuSGD at a learning rate of 0.01. We note explicitly that this setting *overrides* any user-specified learning rate; learning rate is therefore varied only in the dedicated arm of Sec. 4, and readers reproducing this work should be aware that the corresponding flag is otherwise inert.

Several components of the stock augmentation recipe have no anatomical justification on sector ultrasound: mosaic composites sectors into images that cannot physically occur, horizontal flipping violates the left/right convention of an apical view, and hue and saturation jitter are meaningless on grayscale input. We therefore adopt an echo-aware default — mosaic, mixup, copy-paste and flips disabled; mild rotation ($\pm 10^\circ$), translation, scaling and brightness jitter re-

Table 1: 3 trainings of one identical configuration differing only in random seed. The range bounds the smallest difference that can be credibly attributed to a design choice and is used as that threshold throughout. The standard deviation is given because a range grows with the number of runs, so only the SD may be compared against the spread of a larger set of configurations.

	Dice	HD95 (mm)	missed frames
seed 0	0.9151	3.736	1
seed 1	0.9064	3.871	6
seed 2	0.9125	3.821	2
range	0.0087	0.134	5
SD	0.0045	0.068	2.6

tained. We stress that this recipe is chosen on principle and on cost, not because it measures better: Sec. 4 shows it is statistically indistinguishable from the stock recipe.

3.8. Experimental design

The plan is declared in a single machine-readable file and executed by a resumable runner that builds each derived dataset on demand, trains, evaluates on the held-out split, and appends to a summary table. Configurations are hashed after defaults are applied, so a configuration shared between factors is trained once and reused. Every number reported here is regenerated from the run artefacts by a single script.

Training is deterministic: two independent trainings of one configuration, in separate processes thirty hours apart, produced per-frame Dice identical to machine precision. The variance reported in Sec. 4.1 is therefore attributable to seed choice alone, not to nondeterministic kernels or data ordering.

4. Results

4.1. How large an effect is measurable?

We begin with the noise floor, because it determines how everything that follows may be read. Table 1 reports three trainings of one configuration differing only in random seed.

Two consequences follow. First, HD95 varies by 0.134 mm from the seed alone, so no smaller difference should be attributed to a design choice; we adopt this as the threshold throughout.

Table 2: Polygon encoding ceiling: ground-truth masks converted to YOLO polygons and rasterised back, scored against the untouched native ground truth. These are upper bounds no detector trained on that encoding can exceed. n is the vertex budget per contour ($n=0$ keeps every boundary pixel).

Encoding	n	Dice	HD95 (mm)	Myo. target
Filled contour (ours)	32	0.9920	0.513	0.9953
Filled contour (ours)	96	0.9970	0.308	0.9984
Seam-cut annulus	32	0.9883	0.741	0.9671
Seam-cut annulus	96	0.9968	0.308	0.9939

Second, and more troubling for evaluation practice generally, a paired Wilcoxon test between seeds 0 and 1 — runs identical in every respect except the random seed — returns $p=0.035$ on HD95. A paired test over 200 frames is powerful enough to certify pure seed variation as a significant effect. Reporting $p < 0.05$ from single-seed runs on this benchmark is therefore not sufficient evidence of an improvement.

The most seed-sensitive quantity is not an accuracy metric at all but the number of frames on which a structure is missed entirely, which ranges from 1 to 6 across these three runs. We return to this in Sec. 4.8.

4.2. Encoding ceiling

Table 2 reports what each encoding can express as a function of the vertex budget n .

The filled decomposition is effectively lossless at a modest budget: at $n=96$ it reaches 0.9970 Dice and 0.308 mm HD95. Since the CAMUS pixel spacing is 0.308 mm, that HD95 is exactly one pixel — the discretisation floor. The polygon representation is therefore not the limiting factor for a detector on this task, which is the premise the remainder of the study rests on.

The two encodings are nearly indistinguishable on the recomposed label map, because Eq. 1 re-carves the hole either way. They differ on the raw second-class target: at $n=32$ the filled disc is reproduced at 0.9953 Dice against 0.9671 for the seam-cut annulus, a gap of 0.028. A thin ring must spend its vertex budget on two contours rather than one, so each is described half as finely.

4.3. Thirteen factors, no improvements

Table 3 collects the ablation grid. Read against the 0.134 mm floor, the summary is quickly stated: nothing improves on the stock configuration.

Overlap is saturated; boundary distance is not. Across the 32 distinct configurations trained in this study, Dice spans 0.9035 to 0.9177 and HD95 spans 3.70 to 4.07 mm. Stated as ranges these look comparable, but a range grows with the number of samples drawn and so cannot be set against the three-seed floor of Sec. 4.1. The standard deviation can, because it is not biased by sample size, and on that statistic the two axes separate cleanly. Between-configuration Dice varies *less* than the seed floor (0.0037 against 0.0045, a ratio of 0.83), while between-configuration HD95 varies half again as much (0.099 against 0.068, a ratio of 1.46).

Thirteen design factors therefore move Dice by no more than retraining one configuration under a different seed, and move boundary distance measurably more. This is the central negative result of the paper, and it is stronger than a range comparison would suggest: it is not that the factors fail to clear a threshold, but that on overlap they produce less spread than doing nothing at all. Reporting Dice alone on CAMUS therefore conceals most of what an ablation reveals, and is the single clearest argument for the millimetre-scale native-resolution protocol of Sec. 3.4.

Capacity does not help. Model scale from 2.7 to 27 M parameters, input resolution from 512 to 960 px, and mask-prototype resolution all fail to beat the noise floor. Full-resolution prototypes are significantly *worse* (+0.157 mm, $p=0.008$) at 3.5 $\times$ the training cost, which is consistent with over-parameterising the mask path on 1600 images. Input resolution above 640 px is flat: giving the mask head 240×240 prototypes instead of 160×160 changes nothing.

Nor does more data. Training on the full annotated cardiac cycle — 17,006 frames, 10.6 $\times$ the supervision — does not improve ED/ES accuracy. The additional frames interpolate between the two extremes they are meant to support and are highly correlated with them, so they add little independent information.

Training-side factors do register. Removing augmentation entirely costs 0.318 mm ($p<0.001$), the largest effect measured anywhere in this study — larger than 10 $\times$ the parameters, larger than any resolution change. COCO initialisation is worth 0.183 mm ($p=0.009$) despite grayscale sector ultrasound resembling nothing in COCO, which suggests the transfer is of low-level edge and texture filters that 1600 images cannot supply.

The *recipe*, however, does not matter: our anatomically-motivated augmentation is statistically indistinguishable from the stock natural-image recipe ($p=0.31$), even though the latter includes mosaic compositing and horizontal flipping that we argued are anatomically invalid.

Adding horizontal flipping alone to the echo-aware recipe is harmful (+0.203 mm, $p=0.019$), yet the stock recipe — which also flips — is not. We did not isolate this interaction and report it as unresolved.

Neither the objective nor the optimiser. A boundary-weighted mask loss, which upweights pixels near the contour by a factor $1+\lambda$, fails at every $\lambda \in \{2, 5, 10\}$ and at its mildest setting is significantly harmful (+0.216 mm, $p<0.001$). Scaling the segmentation, classification or distribution-focal loss gains, and substituting AdamW or SGD for the automatically selected optimiser, leave HD95 inside the noise floor in all six arms ($p \geq 0.06$). Doubling the weight on the segmentation loss — the term most directly tied to boundary quality — changes HD95 by 0.001 mm.

4.4. The annulus is a non-problem, and the ceiling is not predictive

Neither encoding nor vertex budget produces a measurable difference: filled versus seam at 96 vertices ($p=0.87$), 96 versus 32 vertices ($p=0.31$), and filled versus seam at 32 vertices ($p=0.34$).

The last of these is a genuine falsification and worth stating plainly. The ceiling analysis predicts a 0.228 mm penalty for the seam encoding at 32 vertices. That exceeds the noise floor and should have been visible. The observed difference is 0.019 mm, with the opposite sign. *The encoding ceiling is a valid upper bound but not a predictor of trained-model differences:* a representational penalty of 0.2 mm is simply invisible when model error is 3.7 mm. Practically, any reasonable encoding suffices, and we recommend the filled decomposition because it is simpler to implement — no seam construction, no winding-order bugs — not because it scores better.

4.5. Comparison with semantic segmentation

Table 4 places the configuration alongside nine baselines on the official split. Averaged over three seeds it reaches 0.9113 mean Dice and 3.809 mm HD95 at 2.7 M parameters, against 0.9121/4.081 mm for TransUNet (102 M) and 0.9099/4.270 mm for nnU-Net (15.6 M).

The two axes do not give the same answer, and the difference is the point of this paper. On boundary distance the detector is the best in the benchmark by 0.271 mm, which is twice the 0.134 mm seed floor of Sec. 4.1 and therefore a difference we are willing to attribute. On overlap it is not distinguishable from the leading baselines: tested over the 50 patients rather than the 200 frames, with seeds averaged rather than the best selected, it separates from five of the nine

baselines and not from TransUNet, nnU-Net, UNet-V1 or UNet-ResNet — and its point estimate is 0.0008 Dice *behind* TransUNet.

We report it this way because the alternative is available and flattering: scoring seed 0 over the 200 frames yields 0.9151 Dice and separation from eight of nine baselines. A paper arguing that single-seed frame-level tests overstate differences cannot then rest its own headline on one.

Two external comparisons are available on the same official split. Against a contour-guided boundary-aware method reporting per-structure Dice of 93.95/88.09/89.15 (mean 90.40), our seed mean is 93.86/87.91/91.62 (mean 91.13): ahead on the left atrium and on the three-class mean, and marginally behind on both ventricular classes. Reporting our best seed instead would have given 94.19/88.60/91.74 and a clean sweep, which is precisely the selection this paper argues against. Against a multi-task framework reporting 0.951 Dice for the LV cavity at ED, our LV-endocardial Dice at ED is 0.9523.

We attach limited weight to either, and note that both are additionally compromised as comparisons: the published figures are single runs, ours are seed means, and neither paper reports the variance that would make the difference interpretable. All the gaps here are of the order of our own seed floor. The more informative reference point is the dataset authors’ own: average inter-observer Dice on CAMUS is 0.899. Every model in Table 4 above 0.90 is already operating beyond the level at which two cardiologists agree.

4.6. Latency

Under a single protocol the detector is *near-invariant to input resolution*: 15.7 ms at 256 px and 15.5 ms at 640 px, a 4% increase for 6.25× the pixels. Every encoder–decoder scales with pixel count over the same range, by factors of four to eight. At matched 640 px the detector is between 1.5× and 23× faster than every baseline; end-to-end inference including pre-processing and native-resolution mask assembly is 11.8 ms.

This is a property of the architecture family rather than a tuning outcome: the detector’s cost is dominated by fixed per-layer overhead rather than by convolution over pixels at this scale, so the resolution one would want for boundary accuracy is nearly free, where for an encoder–decoder it is punitive.

Neither FP16 (16.7 ms) nor ONNX Runtime export (23.5 ms) improved on the PyTorch path. At 2.7 M parameters the conversion and transfer overheads dominate. We report this because the opposite is commonly assumed.

4.7. Ejection fraction

Computed with the official implementation at native resolution, the detector reaches 9.24% MAE at $r=0.756$: within the range spanned by the baselines (6.66–9.74%, $r=0.640$ –0.827) but not leading them. We claim no EF advantage.

The ceiling is more informative than the ranking. Feeding the *ground-truth* masks through the identical pipeline yields 5.44% MAE at $r=0.863$. That is the irreducible gap between mask-derived EF and the clinically recorded value, and it consumes the majority of every model’s error. EF on CAMUS is therefore substantially method-limited rather than segmentation-limited.

A fault in the reference implementation. The official Simpson code selects `find_contours(.)[0]` — the first contour returned, ordered by position rather than by area. Ground truth is single-component, so this never manifests and the fault has survived in released code. Model predictions carry occasional spurious blobs, and because EF requires all four of a patient’s masks, a handful of such frames poisons whole patients.

We quantified this by re-running all nine baselines with the filter disabled. The effect is severe where it occurs and absent elsewhere. On UNet-V1, EF correlation collapses from $r=0.825$ to $r=0.080$ and MAE rises from 6.62% to 11.34% purely by removing the filter; DenseContextUNet moves from 9.38% to 12.88% (r 0.763 to 0.550). The remaining seven baselines are unchanged to the precision we report. Whether the fault bites therefore depends on whether a given model emits spurious components at all, which is not a property one can assume — and a pipeline that never checks will not distinguish the two cases.

We verified that this is specific to native-resolution evaluation. On a 256×256 grid the spurious components are sub-pixel and vanish, and filtering slightly *degrades* the result. Practitioners computing EF from predictions at native resolution must apply a connected-component filter; those working on a resampled grid are protected by accident rather than by design.

4.8. Detection reliability is a separate axis

A detector can decline to emit an instance, leaving a structure with no contour at all — a failure mode a semantic network cannot have, since it assigns every pixel a label. This is invisible to Dice computed over detected frames and dominates Dice computed over all frames. In one comparison an apparent 0.0116 Dice deficit was traced entirely to nine dropped instances;

on the 190 frames where both models detected all three structures they were statistically identical ($p=0.72$).

It is also the most seed-sensitive quantity we measured (Table 1), which means single-run miss counts should not be read as properties of a configuration.

Crucially it is eliminable. Lowering the confidence threshold from the natural-image default of 0.25 to 0.10 removes every missed detection — 0 of 200 frames — while leaving Dice and HD95 unchanged. Where every structure is known a priori to be present, the usual precision–recall trade-off does not apply, and the threshold should be set far below the detection default.

5. Discussion

5.1. What the negative result means

Thirteen factors, roughly fifty training runs, and no improvement on a stock configuration beyond run-to-run variance. It is worth being precise about what this does and does not establish.

It does not establish that CAMUS segmentation is solved, nor that architecture is irrelevant in general. It establishes something narrower and, we think, more useful: on this benchmark, at this data scale, the design choices practitioners routinely tune — capacity, resolution, mask-head resolution, label encoding, augmentation recipe, loss weighting, optimiser — do not move boundary accuracy by more than the seed changes it. A practitioner is better served by running three seeds of a stock configuration than by tuning one.

The factors that did register are informative by contrast. Both are *training-side* and both concern the information available to the model rather than its capacity to represent: augmentation (0.318 mm) and COCO initialisation (0.183 mm). Adding parameters, pixels or prototype resolution consistently did nothing or hurt. On 1600 images the model is regularisation- and information-bound, not capacity-bound.

5.2. Where the remaining error lives

The gap between the achieved 3.809 mm HD95 and the 0.31 mm encoding ceiling is a factor of twelve, and none of our interventions closed any of it. Three independent observations point at the annotations rather than the model.

First, mean Dice already exceeds the inter-observer agreement reported by the dataset’s authors (0.899). Above that level the metric increasingly rewards conformity to one annotator’s habits.

Second, the boundary-weighted loss — which upweights exactly the pixels where the contour lies — was the one intervention that was significantly *harmful*. If the contour were where recoverable signal remained, weighting it should have helped. That it hurts is consistent with the contour being where the reference is least reliable, so that emphasising it amplifies label noise.

Third, the EF analysis reproduces the pattern through an independent clinical endpoint: perfect segmentation yields 5.44% EF MAE, and the best model 6.66%. Most of the error survives perfect masks.

We were unable to test this directly. CAMUS released a multi-observer subset — a second and third cardiologist contoured 50 patients, and the original annotator repeated the task seven months later — but that data is not part of the public distribution. We enumerated the public collection and confirmed it contains only single-observer ground truth. An implementation that would compute observer agreement under our exact protocol is included in the released code for anyone who obtains those contours.

5.3. *Implications for evaluation practice*

Three findings of this study are methodological rather than architectural, and we believe they matter beyond CAMUS.

Single-seed ablations produce false positives. A paired Wilcoxon test over the 200-frame CAMUS test set assigns $p=0.035$ to two runs that differ only in random seed. Small test sets and paired tests combine to make seed noise look significant. Several differences reported in the CAMUS literature are smaller than the 0.134 mm we measure from the seed alone. Reporting a seed-variance estimate should be a minimum standard on benchmarks of this size.

Evaluation faults can hide in released reference code. The official CAMUS EF implementation selects a contour by position rather than area. This is correct on ground truth by construction and therefore invisible to the authors’ own validation, but it reduces EF correlation on model predictions from $r=0.825$ to $r=0.080$ at native resolution. A reference implementation being widely used is not evidence that it behaves correctly on the inputs one is actually feeding it.

Protocols must be stated, not assumed. We found that computing boundary metrics on a resampled grid while retaining native spacing under-reports distances; that volumes computed the same way are physiologically impossible (≈ 10 ml rather than ≈ 108 ml) yet yield plausible EF because

the error partially cancels in the ratio; and that latency figures in the literature are not comparable across papers because hardware differs. Each of these silently changes reported numbers without producing any visible error.

5.4. *Practical recommendations*

For practitioners deploying instance segmentation on cardiac ultrasound:

- Use the filled-contour decomposition for annular structures. Not because it is more accurate — it is not — but because it is simpler and its ceiling is one pixel.
- Set the confidence threshold far below the natural-image default. Where every structure is known to be present, 0.10 removed every missed detection at no accuracy cost.
- Do not tune capacity. Spend the compute on seeds instead.
- Apply a connected-component filter before any volume or EF computation.
- Measure latency yourself, under one protocol, at the resolution you will deploy at.

5.5. *Limitations*

This study covers one dataset, one acquisition protocol and one GPU class, with a single official split rather than cross-validation, so the reported spread reflects seed rather than split variation. Most ablation arms are single-seed and are interpreted only against the measured noise floor, never individually; where an arm approached that floor we have said so rather than claiming an effect. Test-time augmentation is unavailable for segmentation models in the framework used — it is silently ignored — so it is untested rather than found ineffective. Latency is measured on a laptop discrete GPU rather than on the embedded accelerator the deployment argument ultimately points at. Finally, the annotation-limit hypothesis is inferred from three converging observations rather than measured directly, because the multi-observer data is not public.

6. Conclusion

We set out to determine what still improves boundary accuracy on a cardiac ultrasound benchmark whose overlap metric has reached the level of annotator agreement, using real-time instance segmentation as the test bed.

The representational obstacle we expected to matter does not. The CAMUS myocardium is an annulus and an instance mask is a polygon, but encoding the epicardium as a filled contour and recovering the myocardium by subtraction costs one pixel of Hausdorff distance, and no encoding or vertex budget we tested produces a measurable difference in a trained model. The encoding ceiling is a valid upper bound; it is not a predictor.

Nor does anything else. Across thirteen factors and roughly fifty training runs — capacity, generation, input resolution, mask-prototype resolution, augmentation recipe, pretraining, data scale, a boundary-weighted loss, the three loss gains and the optimiser — no configuration change improves on a stock setting once run-to-run variance is measured. The two factors that do register, augmentation and initialisation, both concern the information available to the model rather than its capacity. The resulting configuration reaches 0.9113 seed-mean Dice and 3.809 mm HD95 at 2.7 M parameters: the best boundary accuracy in the benchmark, with overlap indistinguishable from the leading baselines. It is near-invariant to input resolution where encoder-decoders scale with pixel count.

Three findings generalise beyond this architecture. Overlap has saturated on CAMUS, so boundary distance in millimetres is what discriminates methods and reporting Dice alone conceals most of what an ablation reveals. A paired test over a 200-frame test set certifies pure seed variation as significant, so single-seed ablations on benchmarks of this size produce false positives. And the reference implementation for CAMUS ejection fraction selects a contour by position rather than area — harmless on ground truth, ruinous on predictions.

Taken together these point to the annotations as the operative limit on in-distribution accuracy, supported by three converging observations: mean Dice already exceeds inter-observer agreement, weighting the loss toward the contour is actively harmful, and perfect segmentation still leaves most of the ejection fraction error intact. If that reading is right, the productive questions are no longer about architecture on this dataset. They are about generalisation across scanners and centres, about temporal consistency across the cardiac cycle — a supervision signal a per-frame model discards — and about whether the benchmark’s reference standard can support the precision that current methods are already reporting against it.

Data and code availability

CAMUS is publicly available from the original authors. All code, the machine-readable experiment plan, per-run artefacts and the scripts that regenerate every table and figure in this paper are released at **[TODO: repository URL]**.

Relationship to other work by the authors

A preliminary version of this study appeared as a four-page conference submission, which reported the encoding ceiling, the seed noise floor and a subset of the ablation grid. The present paper adds the native-resolution latency protocol, the ejection-fraction analysis against a ground-truth oracle, the connected-component fault in the reference Simpson implementation, the full thirteen-factor grid, and per-patient rather than per-frame significance testing. Results common to both are reported identically.

Declaration of competing interest

The authors declare no competing interests.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large language model to assist with drafting and copy-editing of the manuscript text, and with implementation and refactoring of the released analysis code. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Acknowledgements

[TODO: funding statement]

References

- [1] S. Leclerc, E. Smistad, J. Pedrosa, et al., Deep learning for segmentation using an open large-scale dataset in 2d echocardiography, *IEEE Transactions on Medical Imaging* 38 (9) (2019) 2198–2210.

- [2] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: Proc. MICCAI, 2015, pp. 234–241.
- [3] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, K. H. Maier-Hein, nnu-net: A self-configuring method for deep learning-based biomedical image segmentation, *Nature Methods* 18 (2) (2021) 203–211.
- [4] J. Chen, Y. Lu, Q. Yu, et al., Transunet: Transformers make strong encoders for medical image segmentation, arXiv preprint arXiv:2102.04306 (2021).
- [5] H. Cao, Y. Wang, J. Chen, et al., Swin-unet: Unet-like pure transformer for medical image segmentation, in: Proc. ECCV Workshops, 2022, pp. 205–218.
- [6] K. He, G. Gkioxari, P. Dollár, R. Girshick, Mask r-cnn, in: Proc. ICCV, 2017, pp. 2961–2969.
- [7] D. Bolya, C. Zhou, F. Xiao, Y. J. Lee, Yolact: Real-time instance segmentation, in: Proc. ICCV, 2019, pp. 9157–9166.
- [8] J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You only look once: Unified, real-time object detection, in: Proc. CVPR, 2016, pp. 779–788.

Table 3: Ablations on the CAMUS test split. Each block varies one factor, all others held at the default configuration (YOLO26n-seg, 640 px, filled-contour encoding, echo-aware augmentation, COCO-pretrained). D: mean Dice; H: HD95 in mm. Best per block in **bold**.

Setting	D	H	Setting	D	H
<i>Annulus encoding</i>			<i>Augmentation recipe</i>		
Filled contour (ours)	0.9151	3.74	Echo-aware	0.9151	3.74
Seam-cut annulus	0.9177	3.70	Stock COCO	0.9121	3.76
<i>Polygon vertex budget</i>			None	0.9064	4.07
96 vertices	0.9151	3.74	Echo-aware + h-flip	0.9095	3.95
32 vertices	0.9105	3.83	<i>COCO pretraining</i>		
<i>Encoding at 32 vtx</i>			COCO-pretrained	0.9151	3.74
Filled, 32 vtx	0.9105	3.83	From scratch	0.9060	3.93
Seam, 32 vtx	0.9119	3.83	<i>Training data</i>		
<i>Model scale / generation</i>			ED/ES only (1.6k)	0.9151	3.74
YOLO26n-seg	0.9151	3.74	+ cycle frames (20k)	0.9054	4.06
YOLO26s-seg	0.9035	3.81	<i>Boundary-weighted loss</i>		
YOLO26m-seg	0.9135	3.82	$\lambda = 0$ (stock)	0.9151	3.74
YOLO11n-seg	0.9109	4.03	$\lambda = 2$	0.9141	3.94
YOLOv8n-seg	0.9137	3.97	$\lambda = 5$	0.9117	3.85
<i>Input resolution</i>			$\lambda = 10$	0.9149	3.78
512 px	0.9104	3.92	<i>Loss gains and optimiser</i>		
640 px	0.9151	3.74	box <i>imes</i> 0.5	0.9153	3.79
800 px	0.9056	3.77	box <i>imes</i> 2	0.9145	3.75
960 px	0.9108	3.81	cls <i>imes</i> 2	0.9095	3.85
<i>Mask prototype resolution</i>			dfl <i>imes</i> 2	0.9120	3.81
1/4 resolution	0.9151	3.74	AdamW	0.9067	3.95
1/2 resolution	0.9117	3.84	SGD	0.9126	3.88
full resolution	0.9118	3.89	<i>Final configuration</i>		
			seed 0	0.9151	3.74
			seed 1	0.9064	3.87
			seed 2	0.9125	3.82

Table 4: CAMUS official test split (50 patients). Dice on the 256×256 grid, HD95 at native resolution in mm. The detector row is the mean over three seeds, not the best of them. Best per column in **bold**. Markers: Holm-corrected paired Wilcoxon over patients against that mean, * $p < 0.05$, † not significant.

Model	Paradigm	Dice	HD95 (mm)	Params (M)
TransUNet †	Hybrid	0.9121	4.08	102.1
YOLO26-seg (ours, mean of 3 seeds)	Detector	0.9113	3.81	2.7
nnU-Net †	CNN	0.9099	4.27	15.6
UNet-V1 †	CNN	0.9085	4.62	31.0
UNet-ResNet †	CNN	0.9081	4.26	29.1
UNet-V2*	CNN	0.8963	5.29	33.0
Swin-UNet*	Transformer	0.8882	5.08	41.8
DeepLabV3+*	CNN	0.8602	5.71	40.3
FPN-UNet*	CNN	0.8596	6.21	32.2
DenseContextU-Net*	CNN	0.8505	8.29	2.2

Table 5: Latency under a single protocol: batch 1, CUDA-synchronised, median of 50 iterations after 10 warm-up, on one RTX 4060 Laptop GPU. Every model is timed at both resolutions so that input size is separated from architecture; no accuracy is claimed at the off-design size. Swin-UNet is shape-constrained and is timed at 224/448.

Model	@256 px	@640 px	ratio
YOLO26n-seg	15.69	15.53	1.0×
UNet-ResNet	5.75	23.50	4.1×
Swin-UNet	17.45	38.48	2.2×
DeepLabV3+	16.90	39.76	2.4×
nnU-Net	10.68	78.45	7.3×
UNet-V1	17.72	119.32	6.7×
UNet-V2	21.43	153.66	7.2×
FPN-UNet	58.40	364.15	6.2×
DenseCtx-UNet	196.71	50163.17	255.0×
<i>YOLO26n-seg, deployment paths</i>			
end-to-end @640		11.75	
FP16 @640		16.69	
ONNX Runtime @640		23.53	

Table 6: Ejection fraction against the clinically recorded value, computed with the official CAMUS biplane Simpson implementation at native resolution with a largest-component filter (Sec. 4.7). The oracle row applies the identical pipeline to the ground-truth masks and bounds what any segmentation can achieve.

Model	MAE (%)	bias (%)	r	EDV (ml)
UNet-V1	6.62	-0.42	0.825	105.4
TransUNet	7.36	-1.20	0.826	105.9
UNet-ResNet	7.42	-0.98	0.807	107.1
UNet-V2	7.56	0.80	0.781	107.9
nnU-Net	8.32	0.76	0.783	106.5
YOLO26n-seg	9.24	-4.08	0.756	105.3
DenseCtx-UNet	9.38	-7.22	0.763	100.6
Swin-UNet	9.44	-1.96	0.640	101.4
DeepLabV3+	9.58	-7.14	0.706	90.4
FPN-UNet	10.60	-8.52	0.704	96.1
<i>oracle (ground-truth masks)</i>	5.44	3.12	0.863	108.0